

EE5239 Nonlinear Optimization HW 6 Cover Sheet

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Student name: _____

- Date assigned: Mon 12/02/2024
- Date due: **Monday 12/16/2024** Midnight
- This cover sheet must be signed and submitted along with the homework answers on additional sheets.
- By submitting this homework with my name affixed above,
 - I acknowledge that I am aware of the University’s policy concerning academic misconduct (appended below),
 - I attest that the work I am submitting for this homework assignment is solely my own, and
 - I understand that suspiciously similar homework submitted by multiple individuals will be reported for investigation.
- Academic Misconduct in any form is in violation of the University’s Student Disciplinary Regulations and will not be tolerated. This includes, but is not limited to: copying or sharing answers on tests or assignments, plagiarism, having someone else do your academic work or working with someone on homework when not permitted to do so by the instructor. Depending on the act, a student could receive an F grade on the test/assignment, F grade for the course, and could be suspended or expelled from the University.

1 Written Problems

1. Consider the following problem

$$\min_x f(x) := \frac{2}{3}|x|^3 + \frac{1}{2}x^2.$$

Consider using the steepest gradient descent with stepsize $\alpha^k = 1/(k+1)$, and suppose $x^0 = 1$. Does the algorithm converge to the global min $x^* = 0$? Please explain your answer.

2. Prove that $x^* = (1, 1/2, -1)$ is optimal for the optimization problem (using KKT conditions)

$$\begin{aligned} \min \quad & x^T P x + 2q^T x + r \\ \text{s.t.} \quad & -1 \leq x_i \leq 1, \quad i = 1, 2, 3 \end{aligned}$$

where

$$P = \begin{bmatrix} 13 & 12 & -2 \\ 12 & 17 & 6 \\ -2 & 6 & 12 \end{bmatrix}, \quad q = \begin{bmatrix} -22.0 \\ -14.5 \\ 13.0 \end{bmatrix}, \quad r = 1.$$

3. Consider a smooth curve on the plane described by the equation

$$h(x) = 0,$$

where $h : \mathbb{R}^2 \rightarrow \mathbb{R}$ is continuously differentiable. Let y and z be two points that lie in relation to the curve as shown in the following figure. Show that if a point x^* minimizes the sum of the Euclidean distances

$$\|y - x\| + \|z - x\|$$

over all points x on the curve, then the angles ϕ_y and ϕ_z shown in the following figure must be equal.

4. Suppose you have a huge data matrix that cannot fit into the memory of a single computer. Suppose you have a set of K computing nodes, and each of them are connected to a central computer (see the figure above).

The question is, can you design a modified version of the Power Method such that the following conditions are satisfied:

- Your algorithm allows the entire data set to be divided into pieces, each piece is stored in one computing node.
- During the execution of the algorithm, there is no exchange of the actual data set among the computing nodes and the central controller (note, this may be because of privacy issue, or simply because exchanging such data matrices are too expensive).

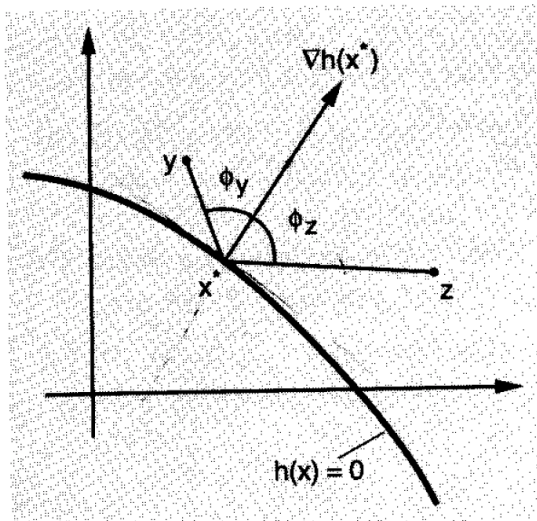
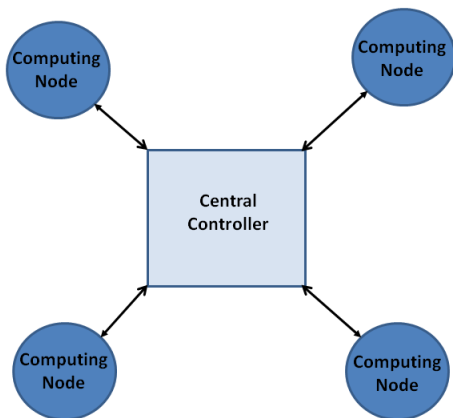


Figure 3.1.5. Fermat's principle; cf. Exercise 3.1.3. The angles ϕ_y, ϕ_z must be equal.



- During the execution of the algorithm, you are allowed to exchange limited information (which can succinctly summarize the data, but such information cannot be the dataset itself) among the nodes and/or the central controller.
- Try to design your algorithm so that at each iteration, the amount of data exchange is as small as possible.

Note: You only need to describe the algorithm, but do not need to implement it.